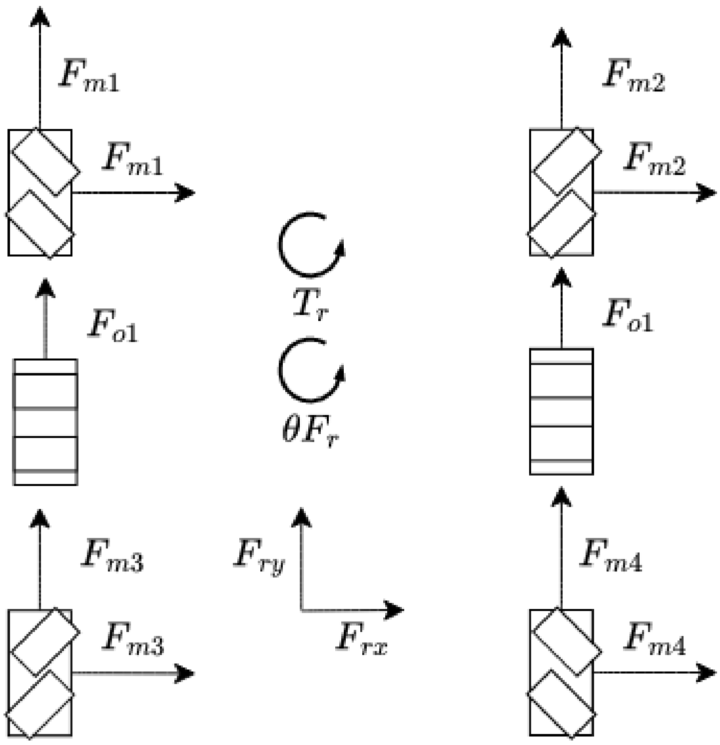

Given

Note: These are the correct formulas for velocity. Incorrectly throughout the proof I used velocity instead of force. Everything in the proof should be replaced with velocity. However as in this case force and velocity are proportional it doesn't change anything

$$v_x = \frac{1}{4}(v_{m1} + v_{m4} - v_{m2} - v_{m3})$$

$$v_y = \frac{1}{6}(v_{m1} + v_{m2} + v_{m3} + v_{m4} + v_{o1} + v_{o2})$$

$$\tan(\theta) = \frac{v_y}{v_x}$$

$$\omega = \frac{L+W}{24}(v_{m2} + v_{m4} - v_{m1} - v_{m3}) + \frac{W}{12}(v_{o2} - v_{o1})$$

$$\frac{L+W}{24}(v_{m2} + v_{m4} - v_{m1} - v_{m3}) = \frac{W}{12}(v_{o2} - v_{o1})$$

The incorrect velocities

$v_{xm} = v_{m1} + v_{m4} - v_{m2} - v_{m3}$, $v_{xo} = 0$

$v_{ym} = v_{m1} + v_{m2} + v_{m3} + v_{m4}$, $v_{yo} = v_{o1} + v_{o2}$

$v_x = v_{xm}$, $v_y = v_{ym} + v_{yo}$, $\tan(\theta) = \frac{v_y}{v_x}$

$$v_x = v_{m1} + v_{m4} - v_{m2} - v_{m3}$$

$$v_y = v_{m1} + v_{m2} + v_{m3} + v_{m4} + v_{o1} + v_{o2}$$

$$\tan(\theta) = \frac{v_y}{v_x}$$

$v_{im} = v_{m2} + v_{m4}$

$v_{lm} = v_{m1} + v_{m3}$

$\omega_m = \frac{L+W}{4}(v_{im} - v_{lm})$

$\omega_o = \frac{W}{2}(v_{o2} - v_{o1})$

$\omega = \omega_m + \omega_o$

$\omega = \frac{L+W}{4}(v_{im} - v_{lm}) + \frac{W}{2}(v_{o2} - v_{o1})$

$$\omega = \frac{L+W}{4}(v_{m2} + v_{m4} - v_{m1} - v_{m3}) + \frac{W}{2}(v_{o2} - v_{o1})$$

$$\frac{L+W}{4}(v_{m2} + v_{m4} - v_{m1} - v_{m3}) = \frac{W}{2}(v_{o2} - v_{o1})$$

$v_r = \sqrt{v_x^2 + v_y^2}$

$T = (m, o)$, $N^m = (1, 2, 3, 4)$ and $N^o = (1, 2)$,

$\forall T \forall N : T_n \in \left[T_{N_{\text{min}}}, T_{N_{\text{max}}}\right]$

$\forall T \forall N : T_n \in \left[T_{N_{\text{min}}}, T_{N_{\text{max}}}\right]$ and $T_{N_{\text{nom}}} = \frac{T_{N_{\text{min}}} + T_{N_{\text{max}}}}{2}$

Find $\text{motor}_{\text{opt}} = \left({}_{(t_n : n \in N, t \in T)}\right)$ st. $\bigcup_{i=1}^{k} \text{motor}_{\text{poss } i} = \text{motor}_{\text{total poss}}$ st. $k \in \mathbb{N}$ st. $\text{motor}_{\text{opt}} \in \text{motor}_{\text{total poss}}$ $\forall N \forall T : T_n \in \text{motor}_{\text{opt}} \in \min_{t,n}\left(\sum_{t \in T} \sum_{n \in N} |t_n - t_{n \text{ nom}}|\right) = S_{\text{abs diff}}$

Proof

let $a \subseteq \mathbb{R}$ and $y \subseteq \mathbb{R}$ if $\min(a) \nleq \min(y)$ and $\max(a) \geq \max(y)$ and $\min(y) \in a$ then $\min(a) = \min(y)$

Therefore if $\forall N \forall T : d_n^T = |T_n - T_{n \text{ nom}}|$ and $\min_{t,n}\left(\sum_{t \in T} \sum_{n \in N} d_n^t\right) = S_{\text{abs diff}}$ then if $u_n^T \geq |T_n - T_{n \text{ nom}}|$ then $\min_{t,n}\left(\sum_{t \in T} \sum_{n \in N} u_n^t\right) = S_{\text{abs diff}}$

Let $\forall N \forall T : d_n^T \geq |T_n - T_{n \text{ nom}}|$

therefore $\forall N \forall T : d_n^T \geq T_n - T_{n \text{ nom}}; d_n^T \geq -(T_n - T_{n \text{ nom}}); d_n^T \geq 0$

therefore $\min_{t,n}\left(\sum_{t \in T} \sum_{n \in N} d_n^t\right) = S_{\text{abs diff}}$

We can then use the simplex algorithm to solve this. To input into the simplex algorithm it says to maximize c^T x subject to $Ax \leq b$ and $x \geq k$ where

- A is an $m \times n$ matrix
- $b \in \mathbb{R}^m$
- $c \in \mathbb{R}^n$
- $x \in \mathbb{R}^n$ are the decision variables
- x is the decision variable vector
- c is the objective-coefficient vector
- $c^T x$ is the objective function
- A is the constraint matrix
- b is the right-hand side vector
- $k \in \mathbb{R}$

$x = \left({}_{((t_n : n \in N, t \in T); (d_n^t : n \in N, t \in T))}\right)$

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{L+W}{4} & \frac{L+W}{4} & -\frac{L+W}{4} & \frac{L+W}{4} & -\frac{W}{2} & \frac{W}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & -1 & -1 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{L+W}{4} & -\frac{L+W}{4} & \frac{L+W}{4} & -\frac{L+W}{4} & \frac{W}{2} & -\frac{W}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{L+W}{4} & \frac{L+W}{4} & -\frac{L+W}{4} & \frac{L+W}{4} & -\frac{W}{2} & \frac{W}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{L+W}{4} & -\frac{L+W}{4} & \frac{L+W}{4} & -\frac{L+W}{4} & \frac{W}{2} & -\frac{W}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ I^6 \mid 0_{6\times 6} \\ -I^6 \mid 0_{6\times 6} \\ -1_{12\times 6} \\ 1_{12\times 6} \\ O^{6\times 6} \mid -1_{6\times 6} \end{pmatrix}$$

$$b = \begin{pmatrix} v_y \\ v_x \\ \omega \\ -v_y \\ -v_x \\ -\omega \\ 0 \\ 0 \\ \forall T \forall N{:}T_{N_{\max}} \\ \forall T \forall N{:}T_{N_{\min}} \\ \forall T \forall N{:}T_{N_{\text{nom}}} \\ \forall T \forall N{:}-T_{N_{\text{nom}}} \\ 0 \end{pmatrix}$$

$$c = \left(0_{6\times 1} \mid -1_{6\times 1}\right)$$

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$$v = A_{1:3}x = \begin{pmatrix} v_x \\ v_y \\ \omega \end{pmatrix}$$

$$v_{xy} = A_{1:2}x = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$$

$$F = \left\{x \in \mathbb{R}^6 : x^{\min} \le x \le x^{\max}, A_3 : x = \omega_{\text{des}}\right\}$$

Given ω_{des} and θ_{des} st. $\omega_{\text{des}} = \omega$ and $\theta_{\text{des}} = \theta$

Find max $v_{r(x)} := \|v_{xy}\|_2 = \sqrt{v_x^2 + v_y^2}$ st. $x \in F$

Proof

For any vector $l \in \mathbb{R}^2$

$\|l\|_2 = \max_{u \in \mathbb{R}^2,\ \|u\|_2 = 1} u^T l$ by Cauchy-Schwarz, for all unit vectors u we have $u^T l \le \|u\|_2\ \|l\|_2 = \|l\|_2$. Equality is achieved by taking $u = \frac{l}{\|l\|_2}$ if $(l \ne 0)$. Hence the max over unit u equals $\|l\|_2$.

$l = v_{xy} = A_{1:2}x$

$\max_{x \in F} \|A_{1:2}x\|_2 = \max_{x \in F} \max_{\|u\|=1} u^T (A_{1:2}x)$ by substitution

$\max_{\|u\| = 1} \max_{x \in F} u^T (A_{1:2}x) = \max_{x \in F} \max_{\|u\|=1} u^T (A_{1:2}x)$ because both F and the unit circle $U = \{u \in \mathbb{R}^2 : \|u\| = 1\}$ are compact and the function $f(u,t) := u^T (A_{1:2}x)$ is continuous on the compact set $U \times F$ and the extreme value theorem guarantees a maximum of f on the product set and both maximizations are finite and over compact sets

$$u^T (A_{1:2}x) = \left(A_{1:2}^T u\right)^T x = c(u)^T x$$

where $c(u) := A_{1:2}^T u \in \mathbb{R}^6$

$\max_{x \in F} \max_{\|u\|=1} u^T (A_{1:2}x) = \max_{\|u\|=1} \max_{x \in F} c(u)^T x$ by substitution

$u = \begin{pmatrix} \cos(\theta_{\text{des}}) \\ \sin(\theta_{\text{des}}) \end{pmatrix}$

$c(u) \coloneqq A_{1:2}^T u = \cos(\theta_{\text{des}}) A_{1:2}^T e_1 + \sin(\theta_{\text{des}}) A_{1:2}^T e_2$

$p \coloneqq \max_{x \in F} c^T x$

$\max_{\|u\|=1} \max_{x \in F} c(u)^T x = \max_{\theta_{\text{des}} \in [0, 2\pi)} p$

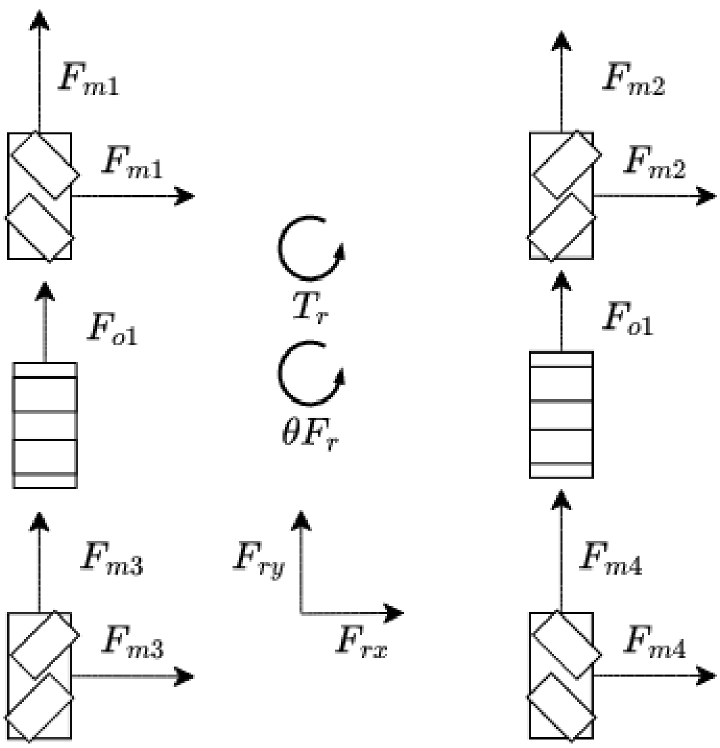
$x_{\text{new}} = x$

$A_{\text{new}} = A_{1:10}$

$b_{\text{new}} = b_{1:10}$

$c_{\text{new}} = \left(\forall x \colon \cos(\theta_{\text{des}}) A_{(\text{new})_x} + \sin(\theta_{\text{des}}) A_{(\text{new})_y} \right)$

This then outputs the optimized vector x_{new}^* we then plug in that vector into v_x and v_y to get v_r



Given

Let

$$x = \begin{pmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ o_1 \\ o_2 \end{pmatrix}$$

and define the chassis velocity map by

$$v_x = \frac{1}{4}(m_1 + m_4 - m_2 - m_3)$$

$$v_y = \frac{1}{6}(m_1 + m_2 + m_3 + m_4 + o_1 + o_2)$$

$$\omega = \frac{L+W}{24}(m_2 + m_4 - m_1 - m_3) + \frac{W}{12}(o_2 - o_1)$$

so that

$$\begin{pmatrix} v_x \\ v_y \\ \omega \end{pmatrix} = Ax$$

where

$$A = \begin{pmatrix} \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ -\frac{L+W}{24} & \frac{L+W}{24} & -\frac{L+W}{24} & \frac{L+W}{24} & -\frac{W}{12} & \frac{W}{12} \end{pmatrix}$$

Let θ_{des} be the desired travel direction and let ω_{des} be the desired angular velocity.

To force motion in the desired direction, write

$$v_x = r\cos(\theta_{\text{des}})$$

$$v_y = r\sin(\theta_{\text{des}})$$

for some $r \geq 0$.

Then maximizing translational speed is equivalent to maximizing r, since

$$r = \sqrt{v_x^2 + v_y^2}$$

Hence the speed-maximization problem is

$$\max r$$

subject to

$$Ax = \begin{pmatrix} r\cos(\theta_{\text{des}}) \\ r\sin(\theta_{\text{des}}) \\ \omega_{\text{des}} \end{pmatrix}$$

and

$$x_i^{\min} \leq x_i \leq x_i^{\max}$$

for all $i \in \{1, 2, 3, 4, 5, 6\}$.

This is a linear program, because both the objective and all constraints are linear in the decision variables x and r.

Proof

Now suppose instead that we want the wheel-speed vector to stay as close as possible to a nominal vector

$$x_{\text{nom}} = \begin{pmatrix} m_{1,\text{nom}} \\ m_{2,\text{nom}} \\ m_{3,\text{nom}} \\ m_{4,\text{nom}} \\ o_{1,\text{nom}} \\ o_{2,\text{nom}} \end{pmatrix}$$

Define auxiliary variables d_i by

$$d_i \geq x_i - x_{i,\text{nom}}$$

$$d_i \geq -(x_i - x_{i,\text{nom}})$$

$$d_i \geq 0$$

Then $d_i \geq |x_i - x_{i,\text{nom}}|$ for each i.

Therefore minimizing

$$\sum_{i=1}^{6} d_i$$

is equivalent to minimizing the total absolute deviation

$$\sum_{i=1}^{6} |x_i - x_{i,\text{nom}}|$$

with the same feasibility constraints as above.

So the full linear program is

$$\min \sum_{i=1}^{6} d_i$$

subject to

$$Ax = \begin{pmatrix} r\cos(\theta_{\text{des}}) \\ r\sin(\theta_{\text{des}}) \\ \omega_{\text{des}} \end{pmatrix}$$

$$x_i^{\min} \leq x_i \leq x_i^{\max}$$

$$d_i \geq x_i - x_{i,\text{nom}}$$

$$d_i \geq -(x_i - x_{i,\text{nom}})$$

$$d_i \geq 0$$

for all $i \in \{1, 2, 3, 4, 5, 6\}$.

At an optimum, each d_i is forced down to the smallest feasible value, so $d_i = |x_i - x_{i,\text{nom}}|$. Thus the objective really is the total absolute deviation from the nominal wheel speeds.

Finally, after solving for x^*, the optimized chassis velocity is

$$\begin{pmatrix} v_x^* \\ v_y^* \\ \omega^* \end{pmatrix} = Ax^*$$

and the translational speed is

$$v_r^* = \sqrt{\left(v_x^*\right)^2 + \left(v_y^*\right)^2} = r^*$$

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